University of BRISTOL

Mathematical Exploration Project

Week 4 · Assessment criteria

MATH30024 · Perspectives in Mathematics · Theme 3B

Learning outcomes of Mathematical Exploration

- Independent learning of mathematics
- Ability to learn from scholarly materials
- Effective group working
- Written communication skills
- Presentation skills

Aims of today

- Understand the **assessment criteria** for the Mathematical Exploration Project.
- **Confirm a project topic** in your group.

How the project is assessed

The project is worth 50% of the unit:

Component	Weight
Written report (group)	25%
Presentation (group)	15%
Individual reflection & use of AI	10%
Total	**50%**

The report and presentation are group marks — everyone in the group receives the same. The reflection is marked individually.

The report (25%) — how those marks split

Criterion	Weight
Breadth and depth of literature studied	20%
Depth of understanding	20%
Explanation and interpretation	20%
Communication	20%
Structure, typesetting and referencing	10%
Teamwork	10%

What the markers look for (1)

Breadth & depth Material not standard in our 2nd/3rd-year curriculum; evidence of substantial independent selection, evaluation and synthesis of scholarly material — in mathematics or in maths education.

Understanding Correct and complete, with evidence of full understanding; does the report explain the depth to which you explored the topic?

Explanation Sophisticated interpretation of complex ideas, consistent and clear; links to familiar material. Could it be understood by a 3rd-year mathematics student?

What the markers look for (2)

Communication Enough introductory material; an attempt to interest the reader — would a 3rd-year want to read to the end?

Structure & referencing Well presented, clear structure, sections linked logically; clear figures and equations; full information in the references, in a consistent style.

Teamwork Does the report read in a single voice, with consistent notation and style, and introductory and concluding sections that bring the parts together as a whole?

If your project has an output

For **outreach** or **computing** projects — a presentation, a display, or code — two of the criteria stretch to cover it:

Depth of understanding *in addition* covers the depth, correctness and suitability of the output itself.

Communication *in addition* covers your description of the output, how you implemented your learning, and — where appropriate — how it was received.

The presentation (15%)

Marked on whether:

- there was detailed **knowledge and understanding** of the main theories and concepts, with answers to all questions demonstrating it;
- the presentation was **well prepared**, with very good presentation skills and appropriate methods;
- there were clear **explanations and interpretations** of complex matters and ideas;
- the team **communicated effectively** to the audience, using a range of visual aids as appropriate;
- the team **worked well together** — all sections at a similar level, consistent, and brought together as a whole.

Individual reflection and use of AI (10%)

Written individually — and it is the one thing that can change your own mark: If you score below 4 out of 10 here, your group mark will be moderated.

- What **you** contributed to the report, and how the group divided the work.
- What you contributed to the group **working well** — organising, reviewing other people's sections, keeping things moving, resolving disagreements.
- One difficulty or decision, and what you learned from it.
- A short, specific **AI-use statement**: which tool, what for, what you kept, and how you checked it. If you used none, say so.

Around **150 words** for the reflection; the AI statement needs only **a few sentences**. Be specific and honest — vague statements score badly, and "I did not use AI" is perfectly acceptable if it is true.

What you may use AI for — Category 3: Selective

You may use AI tools in these three ways only:

1. **Exploring concepts** — as an enhanced form of preliminary study aid. It must not be cited, and you must substantiate any idea you apply by reference to peer-reviewed sources.
2. **Initial research** — to research and explore relevant literature and ideas. You must then read the suggested sources rather than cite from the AI, and responsibility for accuracy remains yours.
3. **Spelling and grammar checkers** — to correct, not to carry out substantial re-writes.

Using AI to draft or rewrite your prose, or to generate mathematics, falls outside this policy.

What a good one looks like

Reflection

I wrote the sections on knot invariants, covering the Jones polynomial and worked examples for the trefoil and figure-eight knots. Beyond my own sections, I set up our shared Overleaf and the reference list, and I read and commented on the section on braid groups — which is how we caught that we were using two different conventions for crossing signs and fixed it early. My main difficulty was deciding how much background to assume: my first draft spent three pages on the Reidemeister moves, which the group judged too much for a third-year reader, so I cut it to one and moved the rest to an appendix. I learned that agreeing the intended reader early saves a great deal of rewriting.

Use of AI

I used an AI assistant to explain the skein relation before I started reading, and to suggest further reading. I cited none of it: I read and cited Adams (2004) and Kauffman (2001) myself. I used a spell-checker for typos, and did not use AI to draft or rewrite anything.

Today: confirm your topic

- Settle your project topic as a group.
- Check it gives everyone a real piece to own.
- Bring any questions to me at the group check-in.

Topic settled by the end of today.

Remember to record your attendance

1. Download the University of Bristol app
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   (or tap on the notification)
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5. Tap ‘Check in’ for this class, then ‘Finish’
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Or go to app.bristol.ac.uk for web check-in

For more information or help, search for ‘record attendance’ on www.bristol.ac.uk

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